1 Label-free node representation learning

Input graph
$G = (V, E, X)$

Node features X

Adjacency A

	v_1	v_2	⋯	v_n
v_1	0	1	⋯	0
v_2	1	0	⋯	1
⋯	⋮	⋮	⋱	⋮
v_n	0	1	⋯	0

Two augmented views

augmentation t_1: edge dropping + feature masking

augmentation t_2: independently sampled

Frozen encoder on the original graph

Shared graph encoder g_ψ (GRACE)

Projection head contrastive training only

Projected $Z^{(1)}$

Projected $Z^{(2)}$

Frozen node representation $\bar{H}$

After pretraining: $\bar{H} = \mathrm{RowNorm}(g_\psi(X, A))$

2 Label-agnostic granular-ball partition

Geometry-only partition in frozen representation space $\bar{H}$

Initialization $B_0 = V$

Recursive 2-means split

Further splitting

Final granular-ball partition in $\bar{H}$

selected ball B_j

Partition all nodes without using labels.

$K \le B_{\max} = 64$

3 Leave-one-out local-consensus targets

(a) Selected granular ball B_j

v

v's own vote removed

(b) Leave-one-out consensus

$p_{j,-v}$ (excluding v)

$$q_{j,-v} = (1-\beta)p_{j,-v} + \beta e_{m_{j,-v}}$$

If the highest vote count is tied, use $q_{j,-v} = p_{j,-v}$.

(c) Construct a soft target t_v

Observed label $(1-\alpha_{j,-v})e_{\tilde{y}_v}$

Local consensus $\alpha_{j,-v}q_{j,-v}$

Soft target t_v

$$t_v = (1-\alpha_{j,-v})e_{\tilde{y}_v} + \alpha_{j,-v}q_{j,-v}$$

$\alpha_{j,-v}$: consensus weight
Fewer than $s_{\min}$ remaining votes: retain the observed label.

4 MLP training and graph-free inference

precomputed soft target t_v

MLP training with GB-LCSS targets

training node $v \in V_L$

node feature x_v

2-layer MLP f_θ

softmax

prediction $\hat{p}_v$

Soft cross-entropy

$\ell_v = -\sum_{c=1}^{C} t_{v,c} \log \hat{p}_{v,c}$

backpropagation: update θ

Graph-free inference (transductive)

test-node feature x_u

Trained MLP f_θ

softmax

class probabilities $\hat{p}_u$

Predicted class: 3

$\hat{y}_u = \arg\max_c \hat{p}_{u,c}$

Observed noisy labels C1 C2 C3 C4 C5

non-voting node

excluded self-vote

data flow

symmetric contrastive loss

Inference input: x_u only; no A, $\bar{H}$, or neighbor access